`Severity` Class Cheat Sheet

m `Severity(sev_name, exp_attachment=None, exp_limit=inf, sev_mean=0, sev_cv=0, sev_a=nan, sev_b=0, sev_loc=0, sev_scale=0, sev_xs=None, sev_ps=None, sev_wt=1, sev_lb=0, sev_ub=inf, sev_conditional=True, name='')`

A `Severity` wraps a frozen `scipy.stats` rv (in `fz`) for a single ground-up loss, optionally cut to a layer by the attachment / limit clause. Mixtures and the `wt` weight are handled by `Aggregate`. The following tables show all m `methods`, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

`name` *object name*
`sev_name` *scipy dist name, or* `(c|d)histogram`
`sev_kind`, `long_name` *resolved labels*
`sev_mean`, `sev_cv` *ground-up mean / CV, or*
`sev_a`, `sev_b` *shape parameters 1, 2*
`sev_loc`, `sev_scale` *shift and scale*
`sev_xs`, `sev_ps` *(d)histogram outcomes / probs*
`sev_wt` *mixture weight (used by* `Aggregate`*)*
`sev_conditional` *losses conditional on attaching the layer (default) or unconditional*
`sev_signed`, `sev_reflect` *allow negative / reflect about 0*
`note`, `hints`, `program` *provenance*

2. Update

`exp_attachment`, `exp_limit` *layer inputs*
`attachment`, `limit`, `detachment` *resolved layer*
`sev_lb`, `sev_ub` *conditional support bounds*
`pattach`, `pdetach` Pr*(attach),* Pr*(detach)*

3. Moments

`sev1`, `sev2`, `sev3` (first three raw moments), m `moms` (m, cv, skew), m `mean`, m `median`, m `std`, m `var`, m `stats`, m `moment`, m `generic_moment`, m `moment_type`, m `support`

4. Statistical functions

m `cdf`, m `sf`, m `pdf`, m `logcdf`, m `logsf`, m `logpdf`, m `ppf`, m `isf`, m `rvs`, m `interval`, m `expect`, m `entropy`, m `vecentropy`, m `nnlf`

5. Validation

None

6. Output dataframes

None

7. Reinsurance

None (layer cut only; see Update)

8. Visualization

m `plot`

9. Risk and pricing

None

10. Approximations

m `fit`, m `fit_loc_scale`, m `freeze`

11. Meta

`fz` (the frozen scipy rv), `shapes`, `numargs` (scipy shape info), `conditional`, `signed`, `bounded`, `tail_class`, `tail_description`, `support_description`, `random_state`, `xtol`

Notes:

A `Severity` models one ground-up severity curve; frequency, mixing, and reinsurance live on `Aggregate`. `sev_mean+sev_cv` or `sev_a(+sev_b)` parameterise the shape; `sev_loc/sev_scale` then shift/scale.

Abbreviations: m=mean, cv=coefficient of variation, sd=standard deviation, var=variance, skew(ness).